
EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 03

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 2](#) and [Note 3A](#)

- (a) How do we deal with piecewise constant inputs as introduced in the notes?
- (b) What conclusions do we get after approximating any function $u(t)$ as being piecewise constant over fixed interval widths Δ ?

2. Simple scalar differential equations driven by an input

In class, you learned that the solution for $t \geq 0$ to the simple scalar first-order differential equation

$$\frac{d}{dt}x(t) = \lambda x(t) \tag{1}$$

with initial condition

$$x(t = 0) = x_0 \tag{2}$$

is given for $t \geq 0$ by

$$x(t) = x_0 e^{\lambda t}. \tag{3}$$

In an earlier homework, you proved that these solutions are *unique*— that is, that $x(t)$ of the form in (3) are the only possible solutions to the equation (1) with the specified initial condition (2).

In this question, we will analyze differential equations with inputs and prove that their solutions are unique.

In particular, we consider the scalar differential equation

$$\frac{d}{dt}x(t) = \lambda x(t) + u(t) \tag{4}$$

where $u(t)$ is a known function of time from $t = 0$ onwards.

- (a) Suppose that you are given an $x_g(t)$ that satisfies both (2) and (4) for $t \geq 0$.

Show that if $y(t)$ also satisfies (2) and (4) for $t \geq 0$, then it must be that $y(t) = x_g(t)$ for all $t \geq 0$.

(HINT: You already used ratios in an earlier HW to prove that two things were necessarily equal. This time, you might want to use differences. Be sure to leverage what you already proved earlier instead of having to redo all that work.)

- (b) Suppose that the given $u(t)$ starts at $t = 0$ (it is zero before that) and is a nicely integrable function (feel free to assume bounded and continuously differentiable with bounded derivative — whatever conditions you assumed in your calculus course when considering integration and the fundamental theorem of calculus). Let

$$x_c(t) = x_0 e^{\lambda t} + \int_0^t u(\tau) e^{\lambda(t-\tau)} d\tau \tag{5}$$

for $t \geq 0$.

Show that the $x_c(t)$ defined in (5) indeed satisfies (4) and (2).

Note: the τ here in (5) is just a dummy variable of integration. We could have used any letter for that local variable. We just used τ because it visually reminds us of t while also looking different. If you think they look too similar in your handwriting, feel free to change the dummy variable of integration to another symbol of your choice.

(HINT: Remember the fundamental theorem of calculus that you proved in your calculus class and manipulate the expression in (5) to get it into a form where you can apply it along with other basic calculus rules.)

(c) **Use the previous part to get an explicit expression for $x_c(t)$ for $t \geq 0$ when $u(t) = e^{st}$ for some constant s , when $s \neq \lambda$ and $t \geq 0$.**

(d) **Similarly, what is $x_c(t)$ for $t \geq 0$ when $u(t) = e^{\lambda t}$ for $t \geq 0$.**

(HINT: Don't worry if this seems too easy.)

3. Uniqueness Counterexample

This problem explores an example of a differential equation that does not have a unique solution. The purpose is to show that uniqueness cannot always be assumed. There is a reason we are making you prove uniqueness to trust solutions.

Along the way, this problem will also show you a heuristic way to guess the solutions to differential equations that is often called “separation of variables.” The advantage of the separation of variables technique is that it can often be helpful in systematically coming up with guesses for nonlinear differential equations. However, as with any technique for guessing, it is not a proof and the guess definitely needs to be checked and uniqueness verified before proceeding.

The idea of separation-of-variables is to pretend that $\frac{d}{dt}x(t) = \frac{dx}{dt}$ is a ratio of quantities rather than what it is — a shorthand for taking the derivative of the function $x(\cdot)$ with respect to its single argument, and then writing the result in terms of the free variable “ t ” for that argument. This little bit of make-believe (sometimes euphemistically called “an abuse of notation”) allows one the freedom to do calculations.

To demonstrate, let’s do this for a case where we know the correct solution: $\frac{d}{dt}x(t) = \lambda x(t)$. This is how a separation-of-variables approach would try to get a guess:

$$\frac{d}{dt}x(t) = \lambda x(t) \tag{6}$$

$$\frac{dx}{dt} = \lambda x \tag{7}$$

$$\frac{dx}{x} = \lambda dt \text{ separating variables to sides} \tag{8}$$

$$\int \frac{dx}{x} = \int \lambda dt \text{ integrating both sides} \tag{9}$$

$$\ln x + C_1 = \lambda t + C_2 \tag{10}$$

$$x(t) = Ke^{\lambda t} \text{ exponentiating both sides and folding constants} \tag{11}$$

With the above guess obtained, $x(t) = Ke^{\lambda t}$ can be plugged in and seen to solve the original differential equation because the steps above are vaguely reversible, if a bit hallucinatory in nature. Then of course, a uniqueness proof is required, but you did that in the previous homework.

To see why this technique is a bit fraught, we will consider the following nonlinear differential equation involving a third root. (If we had more time in making this problem, we would have showed you how this sort of differential equation can arise from a toy physical setting of a inverted pyramidal container that had $x(t)$ liters of water in it, where the rate of water being poured in is proportional to the height of the water $x^{\frac{1}{3}}$. This fractional power arises since volume is a cubic quantity while the water is being poured in at a rate governed by a one-dimensional quantity of length. Similar equations can arise in microfluidic dynamics.)

Anyway, consider the differential equation

$$\frac{d}{dt}x(t) = \alpha x^{\frac{1}{3}}(t) \tag{12}$$

with the initial condition

$$x(0) = 0.$$

Let’s take the “separation-of-variables pill” and see what trip it takes us on:

$$\frac{d}{dt}x(t) = \alpha x^{\frac{1}{3}}(t) \quad (13)$$

$$\frac{dx}{dt} = \alpha x^{\frac{1}{3}} \quad (14)$$

Due to the “abuse of notation,” we can write it as follows

$$x^{-\frac{1}{3}}dx = \alpha dt \quad (15)$$

$$\int x^{-\frac{1}{3}}dx = \int \alpha dt \quad (16)$$

$$\frac{3}{2}x^{\frac{2}{3}} + C_1 = \alpha t + C_2 \quad (17)$$

$$x(t) = \left(\frac{2}{3}\alpha t + C_3\right)^{\frac{3}{2}} \quad (18)$$

That didn’t seem like too wild a ride. Let’s see what rabbit hole we have actually landed in.

- (a) Given our separation-of-variables based calculation, let us guess a solution of the form

$$x(t) = \left(\frac{2}{3}\alpha t + c\right)^{\frac{3}{2}}$$

Show that this is a solution to the differential equation (12), and find the c that satisfies the initial condition.

- (b) Let us guess a second solution:

$$x(t) = 0 \quad (19)$$

Show that this new guess also satisfies (12), and the initial condition ($x(0) = 0$).

- (c) **Show that any solution of the form**

$$x(t) = \begin{cases} 0, & \text{if } t < t_0 \\ \left(\frac{2}{3}\alpha(t - t_0)\right)^{\frac{3}{2}}, & t \geq t_0 \end{cases} \quad (20)$$

also satisfies (12) and the initial condition $x(0) = 0$, for any $t_0 > 0$.

So this actually has *infinitely* many solutions.

- (d) One of the actual sufficient conditions for the uniqueness of solutions to differential equations of the form $\frac{d}{dt}x(t) = f(x(t))$ is that the function $f(x)$ be continuously differentiable with a bounded derivative at the initial condition $x(0)$ and everywhere that the solution purports to go. (You will understand the importance of this condition and where it comes from better when we are in Module 2 of 16B.)

Does this differential equation problem satisfy this condition that would let us trust guessing and checking?

- (e) In this particular case, there is a little bit of a warning in the separation-of-variables trip (you can consider it a dream sequence if you’d like — it has a logic to it, just not a logic that you can rely upon in the real world). **Explain why (15) might be a bit problematic in this case.**
- (f) To see that separation of variables can actually give you reasonable guesses even for equations with interesting solutions, try to use it to solve the following differential equation $\frac{d}{dt}x(t) = \frac{1}{2}x^2$ for initial condition $x(0) = 1$. **Sketch your solution. Describe anything interesting that happens. Does this problem satisfy the uniqueness conditions for $t \in [0, 1]$?**

4. Op-Amp Stability

In this question we will revisit the basic op-amp model that was introduced in EECS 16A and we will add a capacitance C_{out} to make the model more realistic (refer to figure 1). Now that we have the tools to do so, we will study the behavior of the op-amp in positive and negative feedback (refer to figure 2). Furthermore, we will begin looking at the integrator circuit (refer to figure 3) to see how a capacitor in the negative feedback can behave. In the next homework, you will see why it ends up being close to an integrator.

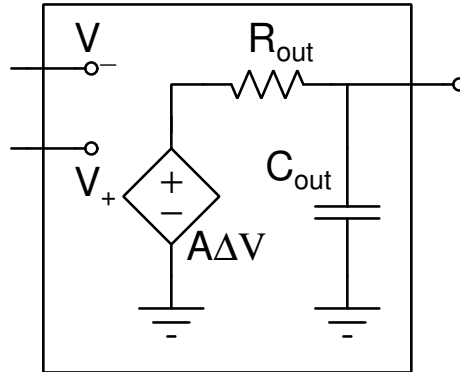


Figure 1: Op-amp model: $\Delta V = V_+ - V_-$

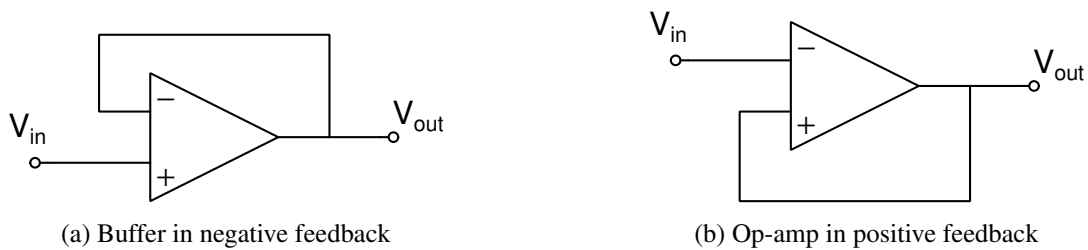


Figure 2: Op-amp in positive and negative feedback

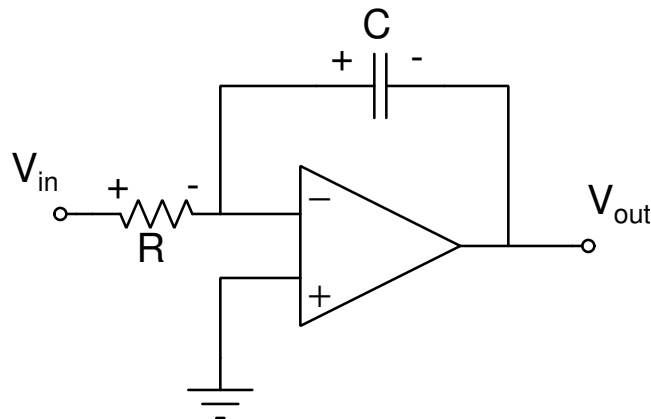


Figure 3: Integrator circuit

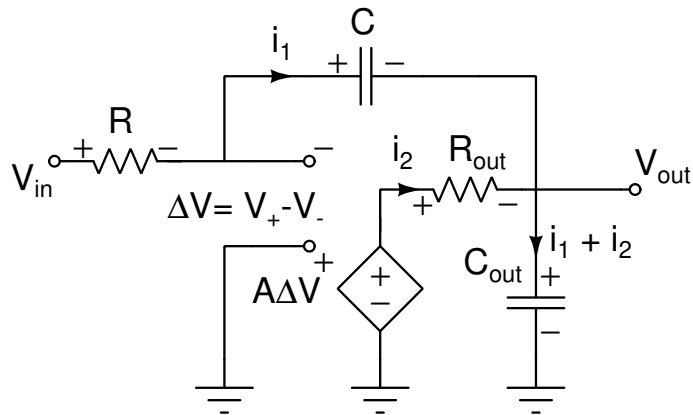


Figure 4: Integrator circuit with Op-amp model

- (a) Using the op-amp model in figure 1 and the buffer in negative-feedback configuration in figure 2(a), **draw a combined circuit**. Remember that $\Delta V = V_+ - V_-$, the voltage difference between the positive and negative labeled input terminals of the op-amp.
(*HINT: Look at figure 4 to see how this was done for the integrator. That might help.*)
Note: here, we have used the equivalent model for the op-amp gain. In more advanced analog circuits courses, it is traditional to use a controlled current source with a resistor in parallel instead.
- (b) Let's look at the op-amp in negative feedback. From our discussions in EECS 16A, we know that the buffer in figure 2(a) should work with $V_{out} \approx V_{in}$ by the golden rules. **Write a differential-equation for V_{out} by replacing the op-amp with the given model and show what the solution will be as a function of time for a static V_{in} . What does it converge to as $t \rightarrow \infty$?** Note: We assume the gain $A > 1$ for all parts of the question.
- (c) Next, let's look at the op-amp in positive feedback. We know that the configuration given in figure 2(b) is unstable and V_{out} will just rail. **Again, using the op-amp model in figure 1, show that V_{out} does not converge and hence the output will rail. For positive DC input $V_{in} > 0$, will V_{out} rail to the positive or negative side? Explain.**
- (d) For an ideal op-amp, we can assume that it has an infinite gain, i.e., $A \rightarrow \infty$. **Under these assumptions, show that the op-amp in negative feedback behaves as an ideal buffer, i.e., $V_{out} = V_{in}$.**
- (e) Let's extend our analysis to the integrator circuit shown in figure 3. Simplifying all the equations, we get a system of differential equations in two variables V_C and V_{out} , where V_C and V_{out} are the voltage drops across the capacitors C and C_{out} . **Fill in the missing term in the following matrix differential equation.**

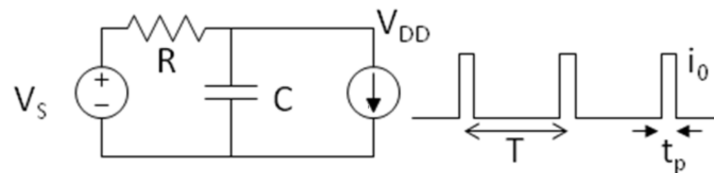
$$\frac{d}{dt} \begin{bmatrix} V_{out}(t) \\ V_C(t) \end{bmatrix} = \begin{bmatrix} -\left(\frac{A+1}{R_{out}C_{out}} + \frac{1}{RC_{out}}\right) & -\left(\frac{1}{RC_{out}} + \frac{A}{R_{out}C_{out}}\right) \\ -\frac{1}{RC} & ? \end{bmatrix} \begin{bmatrix} V_{out}(t) \\ V_C(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{RC_{out}} \\ \frac{1}{RC} \end{bmatrix} V_{in}(t) \quad (21)$$

(*HINT: We picked an easier term to hide. You don't have to write out all the equations and do a lot of algebra to figure out what the missing term is.*)

5. IC Power Supply

Digital integrated circuits (ICs) often have very non-uniform current requirements which can cause voltage noise on the supply lines. If one IC is adding a lot of noise to the supply line, it can affect the performance of other ICs that use the same power supply, which can hinder performance of the entire device. For this reason, it is important to take measures to mitigate, or “smooth out”, the power supply noise that each IC creates. A common way of doing this is to add a “supply capacitor” between each IC and the power supply. (If you look at a circuit board, and the supply capacitor is the small capacitor next to each IC.)

Here’s a simple model for a power supply and digital circuit:

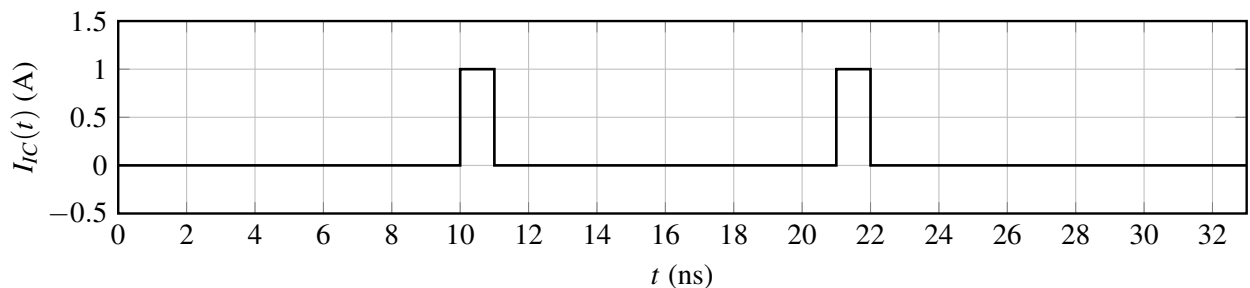


The current source is modeling the “spiky,” non-uniform nature of digital circuit current consumption. The resistor represents the sum of the source resistance of the supply and any wiring resistance between the supply and the load.

The capacitor is added to try to minimize the noise on V_{DD} . Assuming that $V_s = 3\text{V}$, $R = 1\Omega$, $i_0 = 1\text{A}$, $T = 10\text{ns}$, and $t_p = 1\text{ns}$.

- (a) Sketch the voltage V_{DD} vs. time for two periods assuming that $C = 0$. Assume the transient current drawn by IC behaves as follows.

Sketch of transient current drawn by IC



- (b) Give expressions for and sketch the voltage V_{DD} vs. time for two periods for each of three different capacitor values for C : 1pF , 1nF , $1\mu\text{F}$. ($1\text{pF} = 10^{-12}\text{F}$, $1\text{nF} = 10^{-9}\text{F}$, $1\mu\text{F} = 10^{-6}\text{F}$). You may assume $I_C = 0$ for a long time before $t = 0$. Feel free to use the script in the iPython notebook for plotting.
- (c) Launch the attached Jupyter notebook to interact with a simulated version of this IC power supply. Try to simulate the scenarios outlined in the previous parts. For one of these scenarios, keep the RC time constant fixed, but vary the relative value of R vs. C (e.g. compare $R = 1, C = 2 \times 10^{-9}$ to the case where $R = 2, C = 1 \times 10^{-9}$. You may use scientific notations $2\text{e-}9$ and $1\text{e-}9$ in the iPython notebook). **Is it better to have a lower R or lower C value for a fixed RC time constant when attempting to minimize supply noise? Give an intuitive explanation for why this might be the case.**

6. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Anant Sahai.
- Nikhil Shinde.
- Sidney Buchbinder.
- Pavan Bhargava.
- Aditya Arun.
- Gaoyue Zhou.
- Ayan Biswas.
- Sanjit Batra.
- Kris Pister.
- Alex Devonport.